

CAPSTONE PROJECT- Application Packaging

Application packaging is a vital process in modern IT environments, enabling organizations to efficiently deploy, update, and manage software across multiple systems. For this capstone project, you will gain hands-on experience in creating, testing, and deploying application packages using industry-standard tools and best practices. The project is designed to simulate real-world scenarios, guiding you from setting up a clean virtual machine environment to producing a fully functional, customized installer ready for enterprise deployment.

Throughout this process, you will not only develop technical skills in using packaging software and scripting for automation, but will also learn the importance of maintaining standardized, reliable, and secure application deployment methods. By completing each task, you will build a comprehensive understanding of the end-to-end application packaging lifecycle, preparing you for roles in desktop engineering, IT support, and systems administration.

Here's a detailed breakdown for each of the 12 tasks that you are supposed to perform as part of Capstone project in your lab environment and provide the documentation as outcomes of your tasks.

- **3. Install Packaging Tools**
 - Download Admin Studio from the following location. You can use the trial version of the product. <https://community.flexera.com/s/article/download-adminstudio>
 - Install the tool on the VM, following the vendor's installation instructions.
 - Configure basic settings, such as default project directories and licensing, to streamline future packaging tasks.
- **4. Capture an Application Installation Using MSI**
 - Open your packaging tool and start a new project or capture session.
 - Run the application installer while the tool monitors the changes made to the system.
 - Review the captured files, registry entries, and shortcuts, then generate the MSI package from the captured data.
- **5. Create a Silent Installation Script**
 - Identify the silent installation command-line switches for the MSI (e.g., */quiet*, */norestart*).
 - Write a batch (.bat) or PowerShell script that includes these switches for automated deployment.
 - Test the script on a clean VM to ensure the application installs without user interaction or errors.
- **6. Modify MSI Properties Using Orca**
 - Open the MSI file in the Orca editor tool. This tool can be downloaded from Microsoft website.
 - Navigate to the Property table and update fields such as INSTALLDIR, ProductName, and Manufacturer as required.
 - Save the changes and validate that the modified MSI behaves as expected during installation.
- **7. Add Custom Registry Entries**
 - Determine the registry keys and values needed by the application (e.g., license keys, configuration settings).
 - Use the packaging tool to add these entries to the MSI, or manually add them using regedit if required.
 - After installation, verify the registry changes using regedit to ensure accuracy.
- **8. Package an Application with Prerequisites**
 - Identify all required dependencies such as .NET Framework, Visual C++ Redistributables, or other runtime components.
 - Configure your packaging tool to include these prerequisites, ensuring they install before the main application.
 - Test the complete installation workflow to verify all dependencies are installed and the application runs correctly.

- **9. Create a Transform (.MST) File**

- Use the packaging tool or Orca to create a new MST transform file based on the original MSI.
- Customize installation options, properties, or features within the MST without modifying the source MSI.
- Deploy the MSI with the transform by using the command: `msiexec /i app.msi TRANSFORMS=custom.mst`.

- **10. Implement Version Control**

- Assign version numbers to each released MSI package and update the ProductVersion property.
- Document all changes and revisions in a version log or changelog file for future reference.
- Maintain previous versions in a secure repository for rollback or troubleshooting.

- **11. Test Package Installation on a Clean VM**

- Revert the VM to the clean snapshot created earlier.
- Install the newly packaged MSI and monitor the process for errors or warnings.
- Verify that the application launches and functions as intended, with all required files and settings in place.

- **12. Create Uninstall Script**

- Write a script (batch or PowerShell) that calls the MSI uninstall command with silent switches (e.g., `msiexec /x app.msi /quiet`).
- Test the script on the VM to ensure it completely removes the application and associated files/registry entries.
- Document the uninstall process for future reference, especially for enterprise deployments.